NRT Team Project Interim Report

PALM: Physics-Aware Leakage Minimizer

Rodrigo Pires Ferreira, Ray Zhu

Apr 30, 2026

Project Overview

PALM (Physics-Aware Leakage Minimizer) creates train/test splits for scientific ML datasets with reduced data leakage. When structurally similar entities appear in both sets, reported metrics overestimate generalization. PALM builds on DataSAIL [3] and extends it with domain-specific featurization, adaptive distance metrics, 17+ format support, quality metrics, and a web application for molecules, biomolecules, materials, and genes. A parallel effort investigates hypergraph-based partitioning as an alternative backend.

Objectives

1. Develop a unified splitting framework for four scientific entity types.
2. Implement domain-specific feature extraction: fingerprints, physicochemical descriptors, composition features, and language-model embeddings.
3. Automatically select distance metrics and dimensionality reduction based on feature characteristics.
4. Support 17+ scientific input formats with format-preserving output and ML-ready exports.
5. Provide built-in split quality metrics (nearest-neighbor leakage, distribution shift, coverage).
6. Deliver both a CLI and an interactive web application for accessibility.
7. Explore hypergraph-based partitioning as an alternative backend. *(Led by R. Ferreira.)*

Progress to Date

Core pipeline (complete). The end-to-end pipeline—data loading, feature extraction, adaptive distance computation, and splitting—is implemented for both 1D and 2D datasets with random, identity-based, cluster-based, and scaffold splitting techniques.

Featurization (complete). Feature pipelines cover all four entity types with multiple feature sets each (e.g., Morgan fingerprints for molecules; MAGPIE descriptors for materials; ESM2 embeddings for proteins; k-mer frequencies for genes). Results are cached.

Quality metrics and web application (complete). Each run produces nearest-neighbor separation, distribution shift, coverage, and entity overlap metrics with t-SNE visualization. A FastAPI-based web interface provides guided upload, automatic entity-type detection, real-time progress, and interactive dashboards.

Validation (complete). Experiments on molecular and materials benchmarks [1, 4, 2] confirm that cluster-based splitting produces measurably higher test–train separation than random splitting.

Hypergraph splitting (in progress).

Next Steps and Timeline

- **May–Jun:** Integrate hypergraph backend; approximate distance methods for >100K-entity scalability.
- **Jun–Jul:** k-fold cross-validation; domain-specific splitting (protein fold, crystal system).
- **Jul–Aug:** Systematic benchmarking across all entity types; documentation, packaging, and manuscript.

References

[1] J. S. Delaney. ESOL: estimating aqueous solubility directly from molecular structure. *Journal of Chemical Information and Computer Sciences*, 44(3):1000–1005, 2004.

[2] A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, and K. A. Persson. Commentary: The Materials Project: a materials genome approach to accelerating materials innovation. *APL Materials*, 1(1), 2013.

[3] R. Joeres, D. B. Blumenthal, and O. V. Kalinina. Data splitting to avoid information leakage with DataSAIL. *Nature Communications*, 16:3337, 2025.

[4] Z. Wu, B. Ramsundar, E. N. Feinberg, J. Gomes, C. Geniesse, A. S. Pappu, K. Leswing, and V. Pande. MoleculeNet: a benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018.